Sample Midterm Exam 3

Econ 5030

1 Preferences

Consider an agent with preferences over $X = \mathbb{R}^2_+$ defined as follows: For all $x, x' \in X$,

$$x \succsim x' \quad \text{if and only if} \quad x_1 + x_2 \;\geq\; x'_1 + x'_2 - 1.$$

(The agent cannot tell apart bundles whose totals differ by less than one unit.)

1.1 Write the definition of complete preferences and clearly demonstrate that these preferences are (or are not) complete on X.

1.2 Write the definition of transitive preferences and clearly demonstrate that these preferences are (or are not) transitive on X.

1.3 Does a utility function exist that represents these preferences on X? Explain.

1.4 Why are completeness and transitivity called *rationality* axioms? What do they allow us to do?

2 Game Theory

2.1 Timing and Strategies

Take a simple two player game with each player having two actions. Player 1's set of actions is $\{a_1, b_1\}$. Player 2's set of actions is $\{c_2, d_2\}$. The players' payoffs over the four possible action combinations are as follows:

$$\begin{aligned} u_1(a_1, c_2) &= 35 & \qquad u_2(a_1, c_2) &= 25 \\ u_1(a_1, d_2) &= 15 & \qquad u_2(a_1, d_2) &= 30 \\ u_1(b_1, c_2) &= 40 & \qquad u_2(b_1, c_2) &= 10 \\ u_1(b_1, d_2) &= 20 & \qquad u_2(b_1, d_2) &= 20 \end{aligned}$$

Suppose this is a sequential choice game in which player 1 goes first, then player 2 observes player 1's choice and makes her choice.

2.1 Write the Normal Form of the game and construct the Game matrix.

2.2 Find the set of strategies that remain after iterated deletion of strictly dominated strategies.

2.3 Find the set of Nash Equilibrium, draw the Game Tree and find the set of Subgame perfect equilibrium.

2.4 Is the equilibrium outcome Pareto efficient? Explain what Player 2 would gain if she could *commit* to an action in advance.

2.2 Mixed Strategies

Consider the simultaneous game below (payoffs are utilities).

	L	R
U	$4, 0$	$0, 2$
D	$1, 3$	$2, 1$

2.5 Show that this game has no pure strategy Nash Equilibrium. Find the mixed strategy Nash Equilibrium and each player's expected payoff.

2.3 Games with Different Preferences

There are two players and each player i has two actions $\{R_i, Q_i\}$. The money each player i receives for the different action pairs (in a game with player j) is described in the matrix below (*this is not the Game Matrix*):

		$P2$	
		R_2	Q_2
$P1$	R_1	\$6, \$6	\$0, \$10
	Q_1	\$10, \$0	\$1, \$1

The game is sequential. Player 1 picks her action first and then player 2 observes the action of player 1 and player 2 picks his action. We will consider 3 cases of this game.

Case 1: Both players only care about their own money with more money being strictly better than less.

2.6 Create a normal form for this game. Write the game matrix and find the set of Nash Equilibrium. Draw the game tree and find the set of Subgame perfect equilibrium.

Case 2: Both players care about each other in a particular way. Each player only cares about the sum of the two players' money and strictly prefers more total money to less.

2.7 Create a normal form for this game. Write the game matrix and find the set of Nash Equilibrium. Draw the game tree and find the set of Subgame perfect equilibrium.

Case 3: There are 3 players: Player 1, Player 2a and Player 2b. Player 1 only cares about the sum of money of herself and Player 2b, with more money being strictly better. Player 2a only cares about her own money, with more money strictly better. Player 2b only cares about the sum of money of herself and Player 1, with more money being strictly better.

In this case, Player 1 makes her decision without knowing for sure who she is playing. With probability $\mu \in (0, 1)$ she will play against Player 2a and with probability $1 - \mu$ she will play against Player 2b. The player not playing the game (2a or 2b) gets zero money no matter what happens.

2.8 You will need to be a little creative to analyze this game. Create a game tree for this game and use a form of backward induction as your solution concept. Explain clearly the difficulties you faced in finding a way to analyze the game.

3 Conceptual Questions

3.1 Explain clearly the idea of the Independence Axiom using a numerical example of lotteries. Why is it needed for an expected utility representation?

3.2 Define the Normal Form of a game. Explain the idea of designing a game to be *dominant strategy solvable*. Why is this attractive to a designer?

3.3 Order the three solution concepts (dominant strategies, rationalizability / IDSDS, Nash equilibrium) by the strength of their assumptions. Where do the Nash equilibria of a game sit relative to the rationalizable strategies?

3.4 Market games: what does adding production to a pure exchange economy add to the analysis? In an oligopoly, what two kinds of inefficiency can arise?

4 Cournot Duopoly with Asymmetric Costs

Two firms compete in quantities. Market inverse demand is $P(Q) = 120 - Q$, $Q = q_1 + q_2$. Firm 1 has cost $C_1(q_1) = 20q_1$ and firm 2 has cost $C_2(q_2) = 40q_2$.

4.1 Write the Normal Form of this game and each firm's profit function.

4.2 Find each firm's best response function and graph both on the q_1–q_2 plane.

4.3 Find the Cournot Nash equilibrium (q_1^c, q_2^c), Q^c and p^c. Which firm produces more, and why?

4.4 Find each firm's profit, CS, PS and TS.

4.5 Find the efficient outcome: how much should be produced and *by which firm*? Compute the DWL and decompose it into (i) loss from underproduction and (ii) loss from the wrong firm producing. Illustrate on the p–Q plane.

4.6 Suppose firm 1's marginal cost falls from 20 to 14. Find the new equilibrium and explain how each firm's quantity and the price respond.

4.7 For what values of c_2 (holding $c_1 = 20$) does firm 2 produce a positive quantity in equilibrium? What happens otherwise?